PREPARED FOR: J POORNA CHANDAR

Task 74: Count of Smaller Numbers After Self

PROBLEM DESCRIPTION

Given an integer array nums, return an integer array counts where counts[i] is the number of smaller elements to the right of nums[i].

JAVA SOLUTION

```
import java.util.ArrayList;
import java.util.List;

class Solution {
    int[] count;

    public List<Integer> countSmaller(int[] nums) {
        List<Integer> res = new ArrayList<>();
        int n = nums.length;
        count = new int[n];
        int[] indices = new int[n];

        for (int i = 0; i < n; i++) {
            indices[i] = i;
        }

        mergeSort(nums, indices, 0, n - 1);

        for (int i = 0; i < n; i++) {
            res.add(count[i]);
        }
        return res;
    }

    private void mergeSort(int[] nums, int[] indices, int left, int right)
{
        if (left >= right) return;

        int mid = left + (right - left) / 2;
        mergeSort(nums, indices, left, mid);
        mergeSort(nums, indices, mid + 1, right);
        merge(nums, indices, left, mid, right);
    }

    private void merge(int[] nums, int[] indices, int left, int mid, int
right) {
        int[] temp = new int[right - left + 1];
        int i = left, j = mid + 1, k = 0;
        int rightCount = 0;

        while (i <= mid && j <= right) {
            if (nums[indices[j]] < nums[indices[i]]) {
                temp[k++] = indices[j++];
                rightCount++;
```

```
            } else {
                count[indices[i]] += rightCount;
                temp[k++] = indices[i++];
            }
        }

        while (i <= mid) {
            count[indices[i]] += rightCount;
            temp[k++] = indices[i++];
        }
        while (j <= right) {
            temp[k++] = indices[j++];
        }

        for (int p = 0; p < temp.length; p++) {
            indices[left + p] = temp[p];
        }
    }
}
```